

Normal Forms and Complete Relations for Circuits in Agda

Presentations, coset tables, and machine-checked completeness

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Plan for the next 90 minutes

1. Why completeness proofs need a machine 7'
2. A complete worked example: permutation circuits 15'
3. Normal forms from subgroup chains 4'
4. The library, layer by layer 18'
5. A catalogue of verified presentations 10'
6. **New since the paper** 31'
 - scalars and **minimality** of relation sets
 - qupit Clifford circuits, all odd primes
 - **matrix semantics**: $U_n(\mathbb{Z}[\frac{1}{2}, i])$, 2-qubit Clifford+T, Real-Clifford+CH over $\mathbb{Z}[\sqrt{2}]$
 - CNOT-dihedral (phase polynomials); path sums (polynomial cost)
7. Related work, lessons, outlook 5'

How to read the slides

Every frame title carries its provenance:

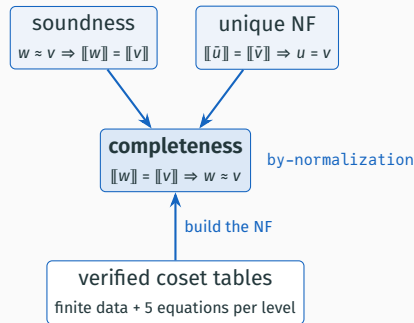
in the paper the POPL'27 submission *Normal Forms and Complete Relations for Circuits in Agda* (tag popl-revision1).

new since the paper developed afterwards on other branches of the same repository (**qupit**, **v-office-Clifford-CH**, **popl**, ...).

All Agda on the slides is typeset by Agda's own $\text{\LaTeX}$ backend from literate files that are scope-checked: against the library on this branch, and against stub declarations for snippets quoted from other branches.

The talk in one slide

- A **circuit** is a word of gates; a **rule set** is a datatype of axioms; the equational theory is the **inductive congruence** they generate.
- A **normal form** is a function on words with a **section** picking canonical representatives.
- **Completeness = soundness + unique normal form**, proved *once*, generically.
- Normal forms are built **one wire at a time** by verified **coset tables**: one “digit” per wire, a mixed-radix numeral.
- Everything is -- safe, axiom-free, setoid-based Agda on the standard library.



Why completeness proofs need a machine

A circuit optimizer, a compiler pass, a diagram normalizer: the justification is always the same — **local rewrite rules** on a syntactic object, preserving a **semantics**.



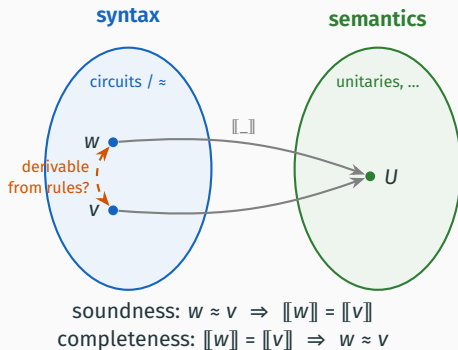
Two questions attach to every rule set:

- **Soundness** — does every rule preserve the semantics?

finitely many small algebraic identities: easy.

- **Completeness** — is every true equation derivable?

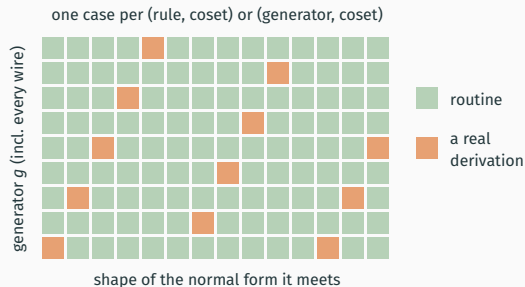
a different animal altogether.



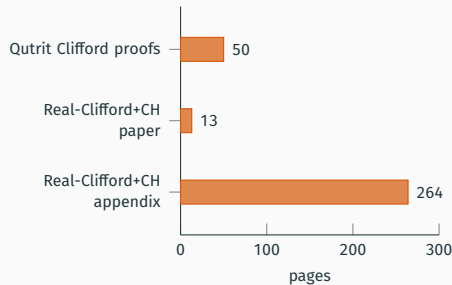
The standard strategy has three steps:

1. define a **normal form** for circuits;
2. show every circuit **rewrites to normal form** using the rules;
3. show **distinct normal forms denote distinct** semantic objects.

Step 2 is where it explodes: for *every* rule and *every* way a generator can meet a normal form, the combination must reduce again — mechanical, unforgiving, enormous.



- Selinger (2015): n -qubit Clifford — a substantial rewrite system and normal-form analysis.
- Li, Mosca, Ross, van de Wetering, Zhao (2025): qutrit Clifford, first complete fragment in odd prime dimension — > 50 pages of proofs.
- Clément, Heurtel, Mansfield, Perdrix, Valiron (2023, 2024): complete theory for full quantum circuits, and its minimal refinement.
- Clément (2026): Real-Clifford+CH — a 13-page paper with an appendix of 264 pages of equational derivations.
- Rig-categorical completeness (POPL '22, '24, '26); Blake (2026) simplifies the rule sets of six near-Clifford fragments.



Long, regular, mechanical: exactly the proofs a proof assistant should check — it does not tire of case splits, and where a case is a computation, its evaluator can carry it out.

- Circuit calculi *are* programming languages; completeness of their equational theory is a PL question.
- It licenses **normal-form decision procedures**, and tells an optimizer author that **no valid rewrite is missing**.
- It underlies the rig-categorical completeness theorems for reversible and quantum languages (Choudhury–Karwowski–Sabry POPL’22, Carette–Heunen–Kaarsgaard–Sabry POPL’24, Fang–Heunen–Kaarsgaard POPL’26).
- The contribution is a **design**: objects that both *compute* and *prove*.

Announced up front: setoids, not quotients

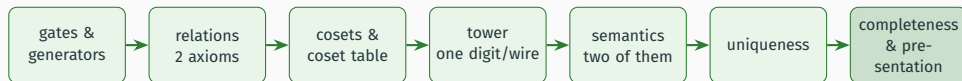
Plain, setoid-based Agda on the standard library — not cubical, not HoTT. A presented monoid is a *setoid on words*, not a quotient type.

- the library **computes with syntax**: coset tables and normal forms pattern-match on concrete representatives, obligations close by evaluation;
- central devices are maps **into** the syntax (sections, coset representatives, amalgamation tails);
- derivations are **data** one can induct on.

0. **A library** for circuit normal forms and complete relation sets: presentations as inductive congruences; normal forms as stdlib function bundles; generic by-normalization; a verified Reidemeister-Schreier engine; solvers by evaluation.
1. **Inductively defined circuits:** wire-indexed generators, structural rules supplied once for every gate set.
2. **A catalogue** of verified presentations: $\mathbb{Z}_n$, S_n , $\mathbb{Z}_N \wr S_n$, Pauli $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$, qubit/qutrit Clifford+T, $U_3(\mathbb{Z}[\frac{1}{2}, i])$.
3. **Monoid-level** amalgamation and Reidemeister-Schreier (after Bian-Selinger).
4. **Presentation theorems** for direct, semidirect, n -fold and amalgamated products, and group extensions.
5. **New stdlib theorems:** semidirect products, amalgamated free products of monoids, extensions, epimorphisms, $\text{Perm}(n)$ as a group.

Every headline theorem is restated with its full type in MainTheorems.agda; --safe, --cubical-compatible, no postulates, no holes, no classical axioms.

A complete worked example: permutation circuits



Circuits built from **one** two-wire gate, the swap σ . They present the **symmetric groups** S_n , at every width n at once.

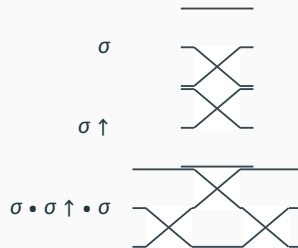
Every later case study is “the same thing, with richer gates, longer towers, and amalgamations where the chain branches”.

A gate set is a family $\text{Gate} : \mathbb{N} \rightarrow \text{Set}$ (gates by arity).
 The circuit layer turns it into **wire-indexed generators**:

```
data Gate : ℕ → Set where
  σ-gate : Gate 2
data Gen : ℕ → Set where
  gate1 : Gate 1 → Gen (1 + n) -- on the bottom wire
  gate2 : Gate 2 → Gen (2 + n) -- on the bottom two wires
  _↑ : Gen n → Gen (1 + n)      -- one wire higher
```

An n -wire circuit is a word of generators,

$\text{Circuit } n = \text{Word } (\text{Gen } n) : [g]^w, \varepsilon(n \text{ bare wires}), c \bullet c'. \ 2 + n$
 is the pattern for $2 + n$, so $\text{gate}_2 \ \sigma$ -gate lives at every width ≥ 2 .



wires drawn bottom-up (wire 0 at the bottom),
 composition left to right; $c \uparrow = \text{wmap } _ \uparrow c$

Gate-specific relations are an inductive family indexed by the wire count; each axiom is stated **once**, at the bottom of the wires, at every width where it fits:

```
data _SRel, _===_ : (n : ℕ) → WRel (Gen n) where
  order : (2 + n) SRel, σ • σ === ε
  yang-baxter : (3 + n) SRel, σ • σ ↑ • σ === σ ↑ • σ • σ ↑
```

Two thirds of the **Coxeter presentation** of S_n : $\sigma^2 = \varepsilon$ and the braid (Yang–Baxter) relation.

The third Coxeter relation — swaps on *disjoint* wires commute — is not about permutations: every circuit theory needs it.

order



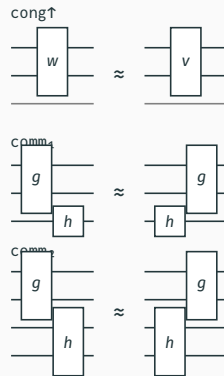
yang-baxter



Circuits are fixed-width slices of a **PROP**. The monoidal structure leaves two traces at fixed width: equations survive the **shift** (tensoring an identity wire), and gates on **disjoint wires commute**.

```
data _VRel, _===_ : (n : ℕ) → CRel n where
  srel : n SRel, w === v → n VRel, w === v
  cong↑ : n VRel, w === v → (1 + n) VRel, w ↑ === v ↑
  comm₁ : (h : Gate 1) (g : Gen n) →
    (1 + n) VRel, [ g ↑ ]w • [ gate₁ h ]w === [ gate₁ h ]w • [ g ↑ ]w
  comm₂ : (h : Gate 2) (g : Gen n) →
    (2 + n) VRel, [ g ↑ ↑ ]w • [ gate₂ h ]w === [ gate₂ h ]w • [ g ↑ ↑ ]w
```

A new circuit theory = a Gate family + its gate-specific relations.
Structural rules and their lemmas come for free.



$\approx$ = the monoid congruence generated by
 $n \text{ VRel}, _ === _;$ deciding it is the word problem.

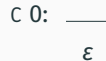
Chain $S_1 < S_2 < \dots < S_n < S_{n+1}$; S_n sits in S_{n+1} as the circuits **not touching the bottom wire**: the circuits $c \uparrow$.

Two circuits are in the same right coset iff they bring the **same wire to the bottom**: $n+1$ cosets.

Representative: the **descending staircase** carrying that wire down.

A coset is a *label*, not a set of circuits:

```
data C : N → Set where
  ε : C n
  σ • _ : C n → C (1 + n)
  [ _ ]c : C n → Circuit (1 + n)
  [ ε ]c = ε
  [ σ • c ]c = σ • ([ c ]c ↑)
```



$$|C\ n| = n + 1 = [S_{n+1} : S_n]$$

Push one generator past a staircase: where do we land, and what is left over on the wires above?

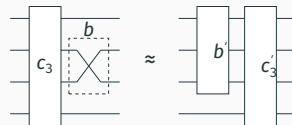
```

ract : C (1+ n) → Gen (2+ n) → Circuit (1+ n) × C (1+ n)
ract ε σ-gen = ε , σ • ε           -- grows
ract (σ • ε) σ-gen = ε , ε         -- cancels (σ² = ε)
ract (σ • σ • c) σ-gen = σ , σ • σ • c -- passes (Yang-Baxter)
ract ε (g 1) = [ g ]w , ε          -- commutes past
ract (σ • c) (g 1) = proj1 (ract c g) ↑ , σ • (proj2 (ract c g))
    
```

Each clause is certified by the **soundness law**, one short derivation per case:

```

ract-sound : ∀ c b → let (b' , c') = ract c b in
    [ c ]c • [ b ]w ≈ b' ↑ • [ c' ]c
    
```



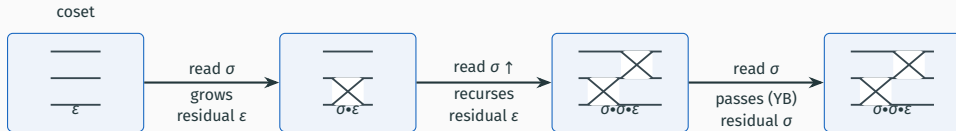
staircase c , then generator b
 $\approx$
 residual b' (one wire up), then staircase c'

The residual is a *circuit*, not a generator:
 in richer gate sets it can be long.



Each step of normalization is an instance of order, of yang-baxter, or of a structural rule (comm_2 , lifted by $\text{cong}\uparrow$). Nothing else is ever used.

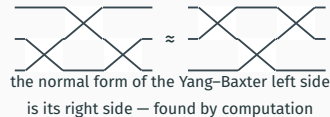
The stateful traversal $_t$ threads the coset left to right and concatenates the residuals — one **level** of normalization:



Level 3 wires: top digit $c_2 = \sigma \cdot \sigma \cdot \epsilon$, residual σ on the upper 2 wires.

Level 2 wires: normalize the residual σ : digit $c_1 = \sigma \cdot \epsilon$.

Rebuild: $[c_1]^c \uparrow \cdot [c_2]^c = \sigma \uparrow \cdot \sigma \cdot \sigma \uparrow$.



Iterating one level per wire is the **coset tower**:

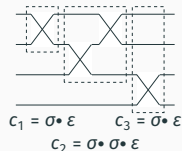
$NF : \mathbb{N} \rightarrow \text{Set}$

$NF 0 = \tau$

$NF (\text{succ } k) = NF\ k \times C\ k \dashv \tau \times C\ 0 \times C\ 1 \times \dots \times C\ k$

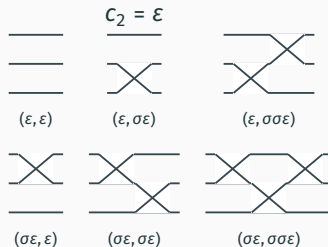
so $|NF\ n| = 1 \cdot 2 \cdots n = n!$: the **factorial number system**.

$nf : \text{Circuit } n \rightarrow NF\ n$ runs the tables; its **section** $inv\text{-}nf$ rebuilds a circuit from the digits — a “staircase of staircases”.



$(tt, c_1, c_2, c_3) \in NF\ 4$

all six normal forms on three wires (S_3):



digit $c_1 \in C\ 1$ (2 values) selects the row, $c_2 \in C\ 2$ (3 values) the column;

section $(c_1, c_2) \mapsto [c_1]^c \uparrow \bullet [c_2]^c$

- **loose:** $\llbracket _ \rrbracket : \text{Circuit } n \rightarrow (\text{Fin } n \rightarrow \text{Fin } n)$, endofunctions — a monoid that is *not* a group;
- **tight:** into $\text{Permutation}'$ -group n , `stdlib` permutations bundled as a `Group` (our `ForStdlib`).

Soundness: small case analysis over the axioms.

Unique NF: $\llbracket \text{inv-nf } u \rrbracket = \llbracket \text{inv-nf } v \rrbracket \Rightarrow u \equiv v$, by induction on the tower: the lower digits fix the last point, so the top digit alone says where it goes.

Completeness then follows generically — even for the *loose* semantics: circuits equal as mere functions are provably equal.

$$\begin{array}{ccc}
 W & \xrightarrow{\llbracket w \rrbracket = \llbracket v \rrbracket} & V \\
 \text{NF law} \parallel & & \parallel \text{NF law} \\
 \text{inv-nf}(\text{nf } w) & \equiv & \text{inv-nf}(\text{nf } v)
 \end{array}$$

soundness: the two normal forms have equal denotations;
uniqueness: so they are *equal*

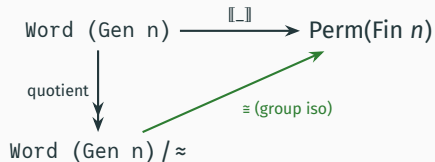
$w \approx \text{inv-nf}(\text{nf } w) = \text{inv-nf}(\text{nf } v) \approx v$ — a derivation assembled from the two normalization runs, i.e. from `order`, `yang-baxter` and the structural rules only.

Soundness, completeness and surjectivity assemble into the **presentation record**: a Grouplike witness (so the presented monoid is a group) and a proof that $\llbracket _ \rrbracket$ is a group isomorphism.

`symmetric-presentation` : $\forall n \rightarrow (n \text{ VRel}, _ == _) \text{ IsPresentationOf } (\text{Permutation}'\text{-group } n)$

Read: the monoid of swap circuits modulo {order, yang-baxter} + structural rules *is* the group of permutations of $\text{Fin } n$ — for every n .

Every presentation theorem in the library has this shape.



gl : the quotient is a group; iso : the induced map is an isomorphism

Normal forms from subgroup chains

$H \leq G$, a **right transversal** picks one $\bar{g}$ per coset Hg :

$$g = h \cdot \bar{g} \quad \text{uniquely,} \quad G \cong H \times (H \backslash G) \text{ (as sets).}$$

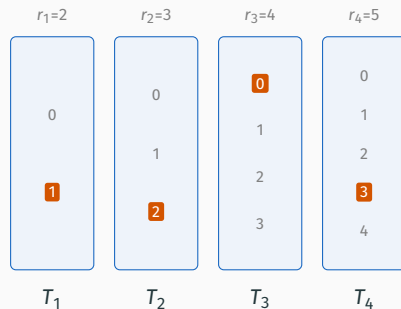
Fix a chain with a transversal T_k for each step:

$$1 = G_0 \leq G_1 \leq \dots \leq G_n = G$$

$$g = t_1 t_2 \dots t_n, \quad t_k \in T_k \quad \text{uniquely.}$$

With indices $r_k = [G_k : G_{k-1}]$: $|G| = r_1 r_2 \dots r_n$ — a **mixed-radix numeral**.

$S_1 < S_2 < \dots < S_n$: radices 2, 3, ..., n = the factorial number system. The conceptual core of **Schreier-Sims**; our chains are chosen *syntactically*.



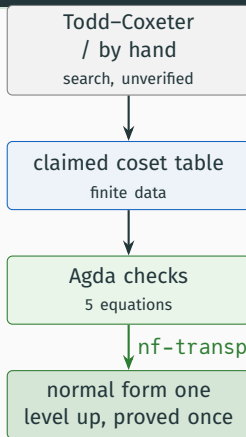
one digit per level: $5! = 120$ elements

Monoid presentation $\langle X \mid R \rangle$: the quotient of X^* by the least congruence $\approx$ containing R ; the **word problem** decides $\approx$. Interpretation $X \rightarrow G$ extends to $X^* \rightarrow G$; *soundness* descends it to $X^*/\approx \rightarrow G$; injective = *complete*.

Given $H \leq G = \langle Y \mid S \rangle$, a **coset table** pushes a generator y past a representative:

$$\bar{t}y = h \cdot \overline{\tau(t, y)}, \quad h \in H.$$

That is exactly *ract*, and the equation is *ract-sound*.

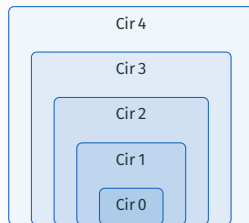


The certificate is separated from the search that found it (pioneered in Agda by Bian–Selinger; here a reusable module).

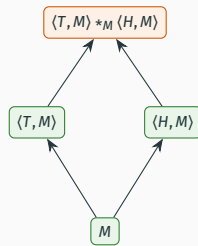
The shift $c \mapsto c \uparrow$ embeds $\text{Cir } k \leq \text{Cir } (k+1)$ as the circuits that do not touch the bottom wire.

So:

- each **wire** adds one digit, ranging over a transversal: staircases for S_n ; Selinger's staircase representatives for Clifford groups; their qutrit analogues;
- a normal form for the whole group is a **tower** — CosetTower;
- where the chain **branches** (two subgroups over a common base, as for Clifford+T), glue by the **amalgamated free product** and its alternating normal form.



tower: nested



branching: amalgam

The library, layer by layer

1. **Syntax is unquotiented**; everything out of it is a function on words — normal forms, coset tables, sections are structurally recursive programs the evaluator runs.
2. **Specifications are stdlib bundles**: a normal form *is* an Injection, RightInverse or Bijection out of the word setoid. No new theory to maintain.
3. **Certificates are finite**: a verified coset table is finite data plus five equations; obligations are case splits whose cases are closed goals.
4. **Indices are chosen for the unifier**: wire counts built from successors only.
5. **Constructions come in pairs**: a syntactic theorem (normal forms combine) and a semantic one (the constructed relation set presents the constructed group).

Examples: S_n , Pauli, Clifford+T, ...

Constructions
 $\times, \rtimes, *_M$, ext.

Circuits
Lift-Relation

Normalization: NF bundles, R-S, cosets, towers

Presentations: $_===_$ and its congruence $_ \approx _$

Words: the free monoid `Word X`

Notations | ForStdlib (semantics)

Agda standard library 2.4

```
data Word (X : Set) : Set where
  [_]w : X → Word X
  ε      : Word X
  _•_    : Word X → Word X → Word X
```

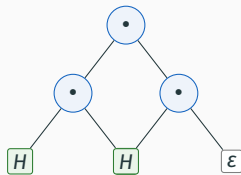
A word remembers its bracketing: terms mirror pen-and-paper calculations, rules apply to any subterm, and associativity is a **congruence generator**, discharged by a solver.

Generator substitutions extend to words (“the unique monoid homomorphism”):

```
_w : (X → Word Y) → (Word X → Word Y)
_w = wconcatmap
```

and a **coset action** extends to words by threading the coset:

```
_t : (C → Y → Word X × C) → (C → Word Y → Word X × C)
```



$(H \bullet S) \bullet (H \bullet \epsilon)$ — a tree;

by-ASSOC flattens both sides to lists and compares by

refl



$(h^t) c (y_1 \bullet y_2) = (w_1 \bullet w_2, c_2)$

One convention everywhere: $_===_$ = the raw axioms Γ ; $_ \approx _$ = the monoid congruence they generate:

```
data _≈_ : WRel X where
  refl : w ≈ w
  sym  : w ≈ v → v ≈ w
  trans : w ≈ v → v ≈ u → w ≈ u
  cong : w ≈ w' → v ≈ v' → w • v ≈ w' • v'
  assoc      : (w • v) • u ≈ w • (v • u)
  left-unit  : ε • w ≈ w
  right-unit : w • ε ≈ w
  axiom      : w === v → w ≈ v
```

- $\approx$ is the word problem of $\langle X \mid \Gamma \rangle$.
- A derivation is a **first-class object**: compute with it, induct on it, translate it (congruence-lifting lemmas are inductions on this type).
- Packaged as the *word setoid* and the presented monoid $\bullet\text{-}\varepsilon\text{-monoid}$; with a Grouplike witness (a left inverse per generator) the group $\bullet\text{-}\varepsilon\text{-group}$.

Contrast: mathlib's `PresentedGroup` (quotient), cubical Agda (HIT).

```
record _IsPresentationOf_ (_===_ : WRel X) (G : Group a ℓ) : Set (a u ℓ) where
  field gl : Grouplike _===_
  module GL = Group-Lemmas _===_ gl
  open GroupMorphisms (Group.rawGroup GL.⋅-ε-group) (Group.rawGroup G)
  field [_] : Word X → Group.Carrier G
      iso : IsGroupIsomorphism [_]
```

- weaker records: `_IsSubPresentationOf_` (monomorphism), monoid versions;
- upgrade lemmas: surjective sub-presentation $\Rightarrow$ presentation; grouplike monoid presentation $\Rightarrow$ group presentation;
- proofs split into soundness, injectivity (by normalization), surjectivity; the records assemble the isomorphism.

Monoid vs group: a real distinction

Words have no formal inverses, so every presentation is at bottom a monoid presentation. $\langle T \mid T^0 = \varepsilon \rangle$ read as a group presentation is $\mathbb{Z}$; as a **monoid** it is $(\mathbb{N}, +, 0)$ — and `Grouplike` is exactly what fails.

From the **word setoid** to a carrier NF , on the nose:

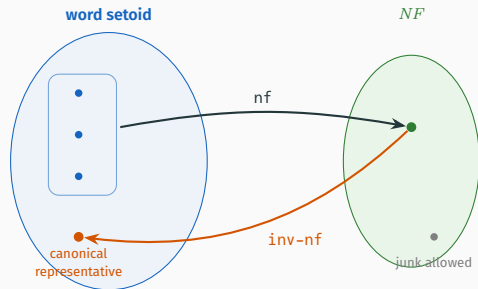
```
NormalFormInjective = Injection
NormalForm          = RightInverse
BijectiveNormalForm = Bijection
```

Injection: $w \approx v \iff \text{nf } w \approx_{NF} \text{nf } v$, and with it

$\text{by-equal-nf} : \text{nf } w \approx_n \text{nf } v \rightarrow w \approx v$

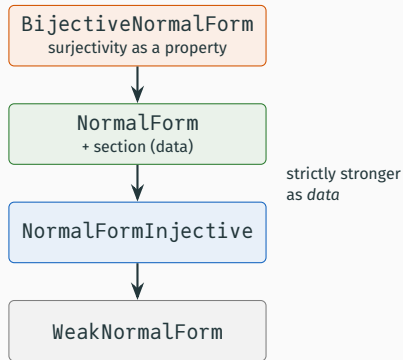
$\approx\text{-dec} : \text{Decidable } \approx_n \rightarrow \text{Decidable } \approx$

prove word equations by **computing**; a decidable carrier **decides the word problem**.



RightInverse adds the **section** inv-nf with $\text{inv-nf} \cdot \text{nf} \approx \text{id}$. Exactness $\text{nf} \cdot \text{inv-nf} \approx \text{id}$ is kept separate.

- Classically, an injective `nf` has a section on its image for free: *choose* a preimage.
- Constructively, choosing is not free: an `Injection` gives **no algorithm** producing a word that realizes a normal form.
- That algorithm is exactly the extra field of `RightInverse` — and it is what completeness consumes.
- So the two are **different strengths** in Agda, though classically equal. Use the weak one for word-problem reasoning, the strong one for completeness and presentations.
- A fourth, crude witness, `WeakNormalForm`: an injective *invariant*, not even congruent.



A NormalForm is **unique** for a semantics $\llbracket _ \rrbracket$ into any setoid when canonical representatives with equal denotations are equal:

```
record UniqueNormalForm (normalForm : NormalForm) : Set (c u d) where
  field unique :  $\forall \{u v : \text{NF}\} \rightarrow \llbracket \text{inv-nf } u \rrbracket \approx_2 \llbracket \text{inv-nf } v \rrbracket \rightarrow u \approx_n v$ 
by-normalization : {normalForm : NormalForm}  $\rightarrow$  UniqueNormalForm normalForm  $\rightarrow$ 
  Congruent  $\approx_1$   $\approx_2$   $\llbracket \_ \rrbracket \rightarrow$  Injective  $\approx_1$   $\approx_2$   $\llbracket \_ \rrbracket$ 
```

Soundness + unique NF $\Rightarrow$ completeness. Five lines of proof; its value: every case study only ever proves unique — a statement about **normal forms**, i.e. concrete finite data, never about arbitrary words.

- by-normalization-wsurj: surjectivity on canonical forms suffices;
- by-completeness: the converse, given an exact section — how uniqueness lifts through products.

$_+_$ recurses on its *first* argument:

$2 + n \rightsquigarrow \text{suc}(\text{suc } n)$, while $n + 2$ is stuck.

So the indices are **left-biased** too:

- $\text{gate}_2 : \text{Gate } 2 \rightarrow \text{Gen } (2 + n)$, not $\text{Gen } k$ with $k \geq 2$;
- k -fold shift: $\text{Gen } n \rightarrow \text{Gen } (k + n)$.

Every index equation the coverage checker meets is solved by **injectivity of $_+_$** , never by arithmetic.

That is why coset tables are written by direct pattern matching.

$\text{lemma-cong}\uparrow$ lifts the *entire* congruence one wire up, by induction over the eight constructors of $\approx$.

left-biased



$\text{Gen } (2 + n) \doteq \text{Gen } (1 + m)$
 $\rightsquigarrow m := 1 + n$ by unification

right-biased



$\text{Gen } (n + 2) \doteq \text{Gen } (1 + m)$
 $\rightsquigarrow n + 2 \doteq \text{suc } m$ stuck:
 a Diophantine equation

```
module SingleLevel ( $\Gamma : \text{WRel } X$ ) ( $\Delta : \text{WRel } Y$ ) ( $C : \text{Set}$ ) ( $I : C$ )
  ( $f : X \rightarrow \text{Word } Y$ ) ( $h : C \rightarrow Y \rightarrow \text{Word } X \times C$ ) ( $[\_ ] : C \rightarrow \text{Word } Y$ )
```

Γ presents a subgroup H of the group G presented by Δ ; C = cosets, I = identity coset, f = embedding of generators, $[_]$ = Schreier section, h = the coset table.

```
module Transfer
```

```
(h=-1f-gen :  $\forall (x : X) \rightarrow ([x]^w, I) \sim ((h^t) I (f x))$ ) -- 1
(h-wd-ax :  $\forall (c : C) \{u t : \text{Word } Y\} \rightarrow u ==_2 t \rightarrow ((h^t) c u) \sim ((h^t) c t)$ ) -- 2
(f-wd-ax :  $\forall \{w v\} \rightarrow w ==_1 v \rightarrow (f^w) w \approx_2 (f^w) v$ ) -- 3
([I]  $\approx_\epsilon [I] \approx_2 \epsilon$ ) -- 4
(h=ract :  $\forall c b \rightarrow \text{let } (b', c') = h c b \text{ in } [c] \bullet [b]^w \approx_2 (f^w) b' \bullet [c']$ ) -- 5
```

(1) the table inverts the embedding at I ; (2) it respects Δ 's axioms, coset by coset; (3) f respects Γ 's axioms; (4) $[I] \approx \epsilon$; (5) the soundness law, as ract -sound.

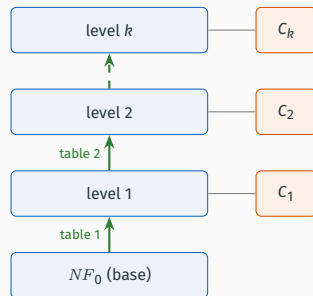
From the five hypotheses, Transfer derives well-definedness of $\text{nf} = (h^{\dagger})I$, the factorization of whole words, injectivity, and the payoff — the **normal-form transport**:

$\text{nf-transp} : \text{NormalForm } \Gamma \text{ } NF \rightarrow \text{NormalForm } \Delta (NF \times C)$

a normal form for the subgroup yields one for the group, **with one more coset digit**.

PackedCosetTable bundles data + hypotheses into one value; CosetTower iterates up an $\mathbb{N}$ -indexed family:

$\text{tower-carrier } NF_0 \ (k+1) = \text{tower-carrier } NF_0 \ k \times C_k$



$\text{carrier } NF_0 \times C_1 \times C_2 \times \dots \times C_k$

Base $\langle \omega \rangle \times \langle S \rangle$ (radices $8 \cdot 4$); adjoin X with $X^2 = \varepsilon$, $(SX)^2 = \omega^2$, ω central. Two cosets, ε and X . The whole table:

```
h : Cr → XSw → Word Sw × Cr
h X-cr X-gen = ε , ε-cr
h X-cr S-gen = (Sw.ω ^ 2 • Sw.S ^ 3) , X-cr
h X-cr ω-gen = Sw.ω , X-cr
h ε-cr X-gen = ε , X-cr
h ε-cr S-gen = Sw.S , ε-cr
h ε-cr ω-gen = Sw.ω , ε-cr
```

from Examples/Amalgamations/CliffordT1.agda (coset type renamed Cr)

Hypothesis (2), every case closed by evaluation:

```
h-wd-ax : ∀ c {u t} → u == t → Set
h-wd-ax X-cr order-ω = by-equal-nf' refl , refl
h-wd-ax X-cr order-S = by-equal-nf' refl , refl
h-wd-ax X-cr order-X = by-equal-nf' refl , refl
h-wd-ax X-cr order-SX = by-equal-nf' refl , refl
h-wd-ax X-cr (comm {X-gen}) = by-equal-nf' refl , refl
h-wd-ax X-cr (comm {S-gen}) = by-equal-nf' refl , refl
h-wd-ax X-cr (comm {ω-gen}) = by-equal-nf' refl , refl
-- ... and the same seven for ε-cr
```

abridged: (by-equal-nf Eq.refl) , Eq.refl in the source

The typechecker runs the table on both sides of the axiom: the cosets coincide and the residues have the same normal form one level down. Qubit Clifford+T: **59** such obligations; qutrit's nine-coset level alone: **72** — every one refl.

Hypothesis (5) at coset X , letter S : the table says $X \cdot S = (\omega^2 S^3) \cdot X$.

The proof, verbatim:

```
lemma-XS : X • S ≈ (ω ^ 2 • S ^ 3) • X
lemma-XS = begin
  X • S ≈⟨ trans (sym right-unit) (sym (cong refl (axiom order-X))) ⟩
  (X • S) • (X • X) ≈⟨ trans (sym left-unit) (sym (cong (axiom order-S) refl)) ⟩
  (S ^ 4) • (X • S) • (X • X) ≈⟨ by-assoc Eq.refl ⟩
  (S ^ 3) • (S • X) ^ 2 • X ≈⟨ cong refl (cong (axiom order-SX) refl) ⟩
  (S ^ 3) • ω ^ 2 • X ≈⟨ sym assoc ⟩
  (S ^ 3 • ω ^ 2) • X ≈⟨ cong (lemma-ω^n 2 (S ^ 3)) refl ⟩
  (ω ^ 2 • S ^ 3) • X ■
```

Division of labour

- **coverage checker**: no case is missed;
- **evaluator**: closes the cases that are computation;
- **human**: writes the few real derivations, which read line by line like the pen-and-paper ones.

(typeset by hand in Agda's colours: this block needs the full module context of `CliffordT1.agda`)

Relation sets over $A \uplus B$ from one primitive: a three-way join whose **mixed** component is the design parameter.

```
data _◇_ (Γ1 : WRel A) (Γ2 : WRel B) (Γ3 : WRel (A ⊔ B))
  : WRel (A ⊔ B) where
  left : Γ1 u v → (Γ1 ◇ Γ2 ◇ Γ3) [u]l [v]l
  right : Γ2 u v → (Γ1 ◇ Γ2 ◇ Γ3) [u]r [v]r
  mid : Γ3 u v → (Γ1 ◇ Γ2 ◇ Γ3) u v
```

Congruence-lifting lemmas (lefts, rights) embed the factors; normal forms transport across.

mixed relation Γ_3	construction
EmptyRel	free product
CommRel	direct product $\times$
ConjRel, ConjRel ^w	semidirect $\rtimes$
AmalgRel $f_1 f_2$	amalgamated $*_M$
SugarRel	definitional extension
$\Gamma \oplus^n$	n -fold power

ConjRel^w: moving h past n conjugates n by a word —
 conj : $H \rightarrow N \rightarrow \text{Word } N$.

Syntactic: normal-form lifting.

- direct / semidirect: carrier $NF_1 \times NF_2$;
- amalgam: a normal form for the *common subgroup* plus one verified coset table per factor:

$$\text{amalNFC } C \ D = (D \wr T) \times \text{List } (C \times D) \times (C \wr T)$$

Semantic: presentation theorems against concrete groups

$$(\Gamma \diamond \Delta \diamond \text{CommRel}) \text{ IsPresentationOf } (G_1 \times G_2)$$

$$(\Gamma \diamond \Delta \diamond \text{ConjRel}^w \text{ conj}) \text{ IsPresentationOf } (G_1 \rtimes G_2)$$

$$(\Gamma \oplus^n) \text{ IsPresentationOf } (\otimes\text{-group } n)$$

$$(P_1 * P_2 \star f_1 \star f_2) \text{ IsPresentationOf } (G_1 *_{G_0} G_2)$$

alternating normal form of an amalgam:



Group **extensions**: given $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ with the conjugation action and the corrections by which lifted relators of Q fail, N 's relations + conjugation rules + corrected lifts present G .

$$1 \longrightarrow N \longrightarrow G \xrightarrow{\text{lift}} Q \longrightarrow 1$$

A presentation theorem cannot be stated without the group it presents — and `stdlib` had the structures but not the **constructions**.

- `Algebra.Construct.SemiDirectProduct`: $N \rtimes H$ via a bundled `Action` (six laws);
- `Algebra.Construct.Amalgamation`: $M *_C N$ of *monoids*, as a setoid — the pushout;
- `Algebra.Construct.Extension`,
 `.SplitExtension`: short exact sequences;
- `Algebra.Morphism.Consequences`: monoid homomorphisms between groups are group homomorphisms;
- `IsMonoidEpimorphism`; `Permutation' n` as a `Group`.

Seven modules, 865 lines — being prepared for upstreaming.

Solvers by evaluation, not reflection

- `by-assoc`: same letter sequence $\Rightarrow w \approx v$;
- `by-passoc`: re-bracket toward a pattern with holes;
- `by-equal-nf`: same computed normal form.

Each maps both sides into a canonical data type by a function proved correct once, then compares by `refl`: the evaluator does the work of a reflective tactic.

A catalogue of verified presentations

	circuit layer	coset tables	tower	$\times, \bowtie$	n -fold power	amalgam	refl- obligations
cyclic $\mathbb{Z}_n$, trivial		✓	base				
symmetric S_n	✓	✓	✓				✓
wreath $\mathbb{Z}_N \wr S_n$	✓			✓	✓		
Pauli $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$				✓	✓		
qubit Clifford+ T		✓	✓	✓		✓	✓
qutrit Clifford+ T		✓	✓	✓		✓	✓
$U_3(\mathbb{Z}[\frac{1}{2}, i])$		✓	✓	✓		✓	✓

base cases

where monoid and group presentations come apart

pure composition

the Pauli group: no coset enumeration at all

stress tests

long towers, twisted relations, branching chains

Relation set	Presents	Index name in MainTheorems
EmptyRel over $\perp$	trivial group	trivial-presentation
TrivialRel over any A	trivial group	trivial-presentation'
$T^{n+1} = \varepsilon$	$\mathbb{Z}/(n+1)\mathbb{Z}$	cyclic-presentation
$T^0 = \varepsilon$	the monoid $(\mathbb{N}, +, 0)$	cyclic-monoid-presentation
order, yang-baxter + structural	permutations of Fin n	symmetric-presentation
$C_p \diamond C_p \diamond \text{CommRel}$	$\mathbb{Z}_p \times \mathbb{Z}_p$	(H-pres)
$(C_p \diamond C_p \diamond \text{CommRel}) \oplus^n$	$(\mathbb{Z}_p \times \mathbb{Z}_p)^n$	pauli-presentation
$(C_N \oplus^n) \diamond S_n\text{-circuits} \diamond \text{ConjRel}^w$	$\mathbb{Z}_N \wr S_n$	wreath-presentation
qubit Clifford+T gate relations	$\cong \text{amalgam } \langle T, M \rangle *_M \langle H, M \rangle$	clifford+T-qubit-isomorphism
qutrit Clifford+T gate relations	$\cong \text{amalgam over } M = \langle S, X, \zeta, H^2 \rangle$	clifford+T-qutrit-isomorphism
$U_3(\mathbb{Z}[\frac{1}{2}, i]), S1-S14$	$\cong (\mathbb{Z}_4 \wr S_3) *_M \langle M, K_{01} \rangle$	U_3 -presentation-isomorphism

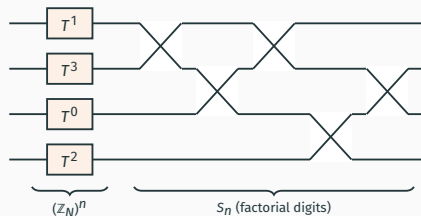
Normal-form carriers: $\text{Fin}(n+1); \mathbb{N}; T \times C1 \times \dots \times C(n-1); NF_1 \times NF_2; NF_0 \times \text{amalg } NFC \ C D; \text{Fin } 8 \times \text{Fin } 4$ (Clifford+T base).

$(\mathbb{Z}_N)^n$ by the n -fold power; S_n acts by permuting coordinates, given **syntactically** as a word-valued conjugation; two finite compatibility lemmas feed the semidirect-product theorem:

wreath-presentation : $\forall n m \rightarrow$
 $((\text{succ } m \text{ Cn, } _ == _) \oplus^n n) \diamond (n \text{ VRel, } _ == _) \diamond \text{ConjRel}^w \text{ conj}$
 $\text{IsPresentationOf } (\text{Wreath.wreath-group } n m)$

- $N = 2$: the hyperoctahedral group B_n of **signed permutations**;
- general N : $G(N, 1, n)$, monomial matrices with N -th roots of unity;
- a family of completeness results over **two naturals**, all at once.

normal form: residues, then staircases



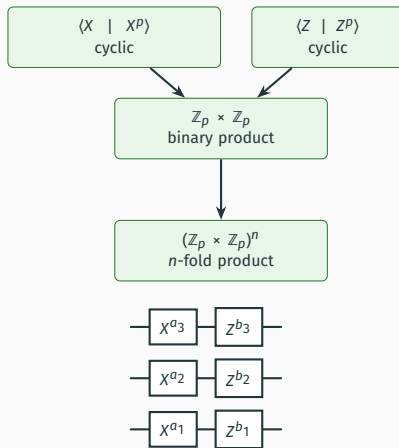
$$\begin{pmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} \in B_4$$

a signed permutation ($N = 2$): one ± 1 per row and column

```

pauli-presentation : ∀ m (prime : Prime (2+m)) n →
  (Γ-Pauli1 m prime ⊞n) IsPresentationOf (Pauli-group m prime n)
  
```

- $p = 2 + m$ prime: a **qubit**;
- $\Gamma\text{-Pauli}_1$: direct-product join of two order- p cyclic presentations (X, Z ; they commute modulo phases);
- n qupits: the n -fold power;
- the proof is **three construction theorems composed** — three lines of Agda, no coset enumeration;
- normal form: $X^{a_1}Z^{b_1} \dots X^{a_n}Z^{b_n}$, $(a_i, b_i) \in \text{Fin } p \times \text{Fin } p$.



Generators $T, H, S, \omega; X := HSSH$:

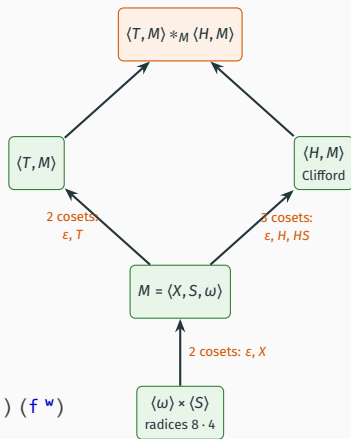
$$\omega^8 = \varepsilon, T^2 = S, S^4 = \varepsilon, H^2 = \varepsilon,$$

$$(SH)^3 = \omega, (TX)^2 = \omega, \omega \text{ central.}$$

Verified structure:

$$\text{Clifford}+T \cong \langle T, M \rangle *_M \langle H, M \rangle, \quad M = \langle X, S, \omega \rangle.$$

```
clifford+T-qubit-isomorphism : IsMonoidIsomorphism
  (rawMonoid (•-ε-monoid _==_)) (rawMonoid (•-ε-monoid amalPres)) (f w)
```



59 well-definedness obligations, every one
(by-equal-nf Eq.refl) , Eq.refl.

With $C = \{T\}$ and $D = \{H, HS\}$, the section of the alternating normal form spells, left to right,

$$M (\varepsilon \mid H \mid HS) (TH \mid THS)^* (\varepsilon \mid T)$$

and read right to left:

$$(\varepsilon \mid T) (HT \mid SHT)^* (\varepsilon \mid H \mid SH) M$$

— the **Matsumoto–Amano normal form** $(\varepsilon \mid T)(HT \mid SHT)^* C$, with its Clifford factor split into a coset representative and an element of M .



The syllable structure falls out of the group theory alone (as observed in the Bian–Selinger line).

Qutrit (ζ a 9th root; X, Z abbreviations):

$$\zeta^9 = \varepsilon, S^3 = \zeta^6, H^4 = \varepsilon, T^3 = Z, (SH)^3 = \varepsilon,$$

$$(THH)^2 = \varepsilon, HHS HHS = SHHS HH, TS = ST,$$

$$TX = \zeta^3 S^2 XT, \zeta \text{ central.}$$

Tower $\langle \zeta \rangle \rightarrow \langle S, \zeta \rangle (3) \rightarrow \langle S, X, \zeta \rangle (9) \rightarrow M (2)$, then two extensions over $M = \langle S, X, \zeta, H^2 \rangle$. Normal form = Glaudell–Ross–Taylor's T -optimal form. The nine-coset level: 72 obligations, all refl.

$U_3(\mathbb{Z}[\frac{1}{2}, i])$ (Bian–Selinger, $n = 3$):

$$U_3(\mathbb{Z}[\frac{1}{2}, i]) \cong (\mathbb{Z}_4 \wr S_3) *_M \langle M, K_{01} \rangle,$$

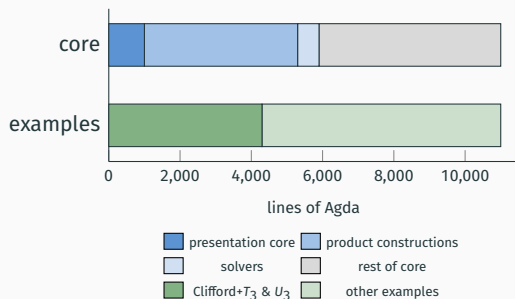
$$M = (\mathbb{Z}_4 \wr S_2) \times \mathbb{Z}_4.$$

The left factor is taken **verbatim** from the wreath development.

$$M(\varepsilon \mid K_{01} \mid K_{01}i_0) \left((X_{12} \mid X_{12}X_{01})(K_{01} \mid K_{01}i_0) \right)^*$$

$$(\varepsilon \mid X_{12} \mid X_{12}X_{01})$$

A **new canonical form** for $U_3(\mathbb{Z}[\frac{1}{2}, i])$ with Matsumoto–Amano-style syllables — for free from the amalgam.



approximate split of $\approx 22\,000$ lines (74 files, Sept. 2026)

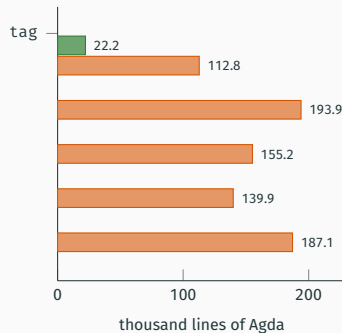
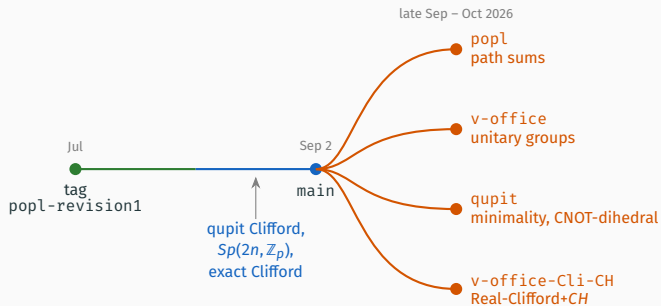
- about **22 000 lines** in 74 files: half reusable core, half examples;
- of 7 000 lines under Presentation: $\approx 1\,000$ core proper, 4 300 the five product constructions (amalgamation alone 1 800), 600 the solvers;
- whole development from scratch, Agda 2.8.0: **1 min 42 s**, peak 3.5 GB; a third of it the amalgamated product (31 s), the three amalgamation studies 19 s, every other module < 4 s.

Limitations stated in the paper: amalgamation studies are syntax-to-syntax (no matrix semantics); continuous-parameter gate sets and infinite-index steps are outside the engine.

New since the paper

What happened after the submission

new since the paper



* .agda outside paper /; 74 files at the tag, 634 on v-office-Cli-CH

paper formalized	gate set / group	semantics	status
Bian, Li, Ross, van de Wetering, Zhao (QPL'26): qubit Clifford, all odd primes	$H, S, CZ; \text{Pauli} \rtimes Sp(2n, \mathbb{Z}_p)$	symplectic maps, Pauli action	complete , every n , every odd prime p
Selinger (LMCS 2015): n -qubit Clifford	ω, H, S, CZ	$Sp(2n, 2) + \text{phases}$	complete (modulo scalars; scalars as extension)
Bian–Selinger (QPL'21): $U_n(\mathbb{Z}[\frac{1}{2}, i])$	two-level X, K, i	matrices over $\mathbb{Z}[\frac{1}{2}, i]$	complete , every n
Greylyn (2014); Bian–Selinger (QPL'22): 2-qubit Clifford+ T	$\omega, H_i, S_i, T_i, CZ$	matrices over $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$	complete ($n \leq 4$; faithful into U_4)
Bian–Selinger (QPL'23): 3-qubit Clifford+CS	S_i, CS, K_i, i	matrices over $\mathbb{Z}[\frac{1}{2}, i]$	in progress (steps 1 and 4 of the R–S chain)
Clément (2026): Real-Clifford+ CH	H, Z, CZ, CH	matrices over $\mathbb{Z}[\sqrt{2}]$	sound; complete on ≤ 1 qubit; ≥ 2 modulo Thm 4.4 (imported by the paper too) + Lemma 8.8 at 3–4 qubits
Amy–Chen–Ross (QPL'17): CNOT-dihedral	$\omega, X, T, CNOT$	phase polynomials	sound; complete on 1 qubit; n modulo imported lemmas
Amy (QPL'18): path sums	$H, S, CZ / H, CNOT, R_k$	exact sums over $\mathbb{Z}[\zeta]$	Clifford equivalence in polynomial cost

plus, in the core: 0-ary gates (scalars), independence of axioms, central and factor-set extensions, generic circuit coset towers

Global phases (ω , ζ , ...) live on no wire, at every width:

```
data Gen : N → Set where
  gate₀ : Gate 0 → Gen n
  gate₁ : Gate 1 → Gen (1 + n)
  gate₂ : Gate 2 → Gen (2 + n)
  _↑ : Gen n → Gen (1 + n)
```

Four structural rules now (qubit, Circuit/Base.agda):

```
data _VRel, _===_ : (n : N) → CRel n where
  srel : n SRel, w === v → n VRel, w === v
  cong↑ : n VRel, w === v → (1 + n) VRel, w ↑ === v ↑
  comm₁ : (h : Gate 1) (g : Gen n) → (1 + n) VRel,
    [ g ↑ ]w • [ gate₁ h ]w === [ gate₁ h ]w • [ g ↑ ]w
  comm₂ : (h : Gate 2) (g : Gen n) → (2 + n) VRel,
    [ g ↑ ↑ ]w • [ gate₂ h ]w === [ gate₂ h ]w • [ g ↑ ↑ ]w
  ω↑=ω : (ω : Gate 0) → (1 + n) VRel, [ gate₀ ω ]w ↑ === [ gate₀ ω ]w
```

There is **no** comm_0 rule: centrality of a scalar is a *theorem*, from the other rules plus one hypothesis — that scalars commute with each other:

```
module Central-Scalars
  (scalars-comm : ∀ {n} (h h' : Gate 0) →
    [ gate₀ {n} h' ]w • [ gate₀ h ]w
    ≈ [ gate₀ h ]w • [ gate₀ h' ]w)
  where
    comm₀ : ∀ (h : Gate 0) (g : Gen n) →
      [ g ]w • [ gate₀ h ]w ≈ [ gate₀ h ]w • [ g ]w
```

(the `let open PB ...` of the source elided)



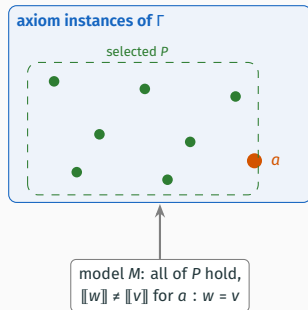
Presentation/Independence.agda (qubit): select some axiom *instances*; is another one derivable from them?

$\text{Select} : \text{WRel } X \rightarrow \text{Set}_1$
 $\text{Select } \Gamma = \forall \{w\ v\} \rightarrow \Gamma\ w\ v \rightarrow \text{Set}$

$_|_ : (\Gamma : \text{WRel } X) \rightarrow \text{Select } \Gamma \rightarrow \text{WRel } X$
 $(\Gamma \mid P)\ w\ v = \Sigma (\Gamma\ w\ v)\ P$

A **model**: a monoid where the selected axioms hold. It separates the instance's two sides $\Rightarrow$ the instance is independent:

$\text{independent} : (a : \Gamma\ w\ v) \rightarrow \neg ([w] \approx^M [v]) \rightarrow \text{Independent } \Gamma\ P\ a$
($_ \approx^M _$ is the monoid's $_ \approx _$ in the source)



soundness extends from axioms to the whole congruence of $\Gamma \mid P$ by StarInterp: a separated instance is not derivable.

Over *any* gate family, for the pure structural theory (Circuit/Independence.agda), each rule fails in a model of the other three:

```
comm1-independent : (h : Gate 1) →
  comm1 h (gate1 {n} h) IndependentOf comm1
ω↑=ω-independent : (ω : Gate 0) → ω↑=ω {n} ω IndependentOf ω↑=ω
cong↑-independent1 : (h : Gate 1) →
  cong↑ (comm1 h (gate1 {n} h)) IndependentOf cong↑
```

and the hypothesis of Central-Scalars is **needed**:

```
scalars-comm-needed : (ω ω' : Gate 0) → ω ≠ ω' →
  ¬ ([ gate0 {n} ω' ]w • [ gate0 ω ]w
    ≈ [ gate0 ω ]w • [ gate0 ω' ]w)
```

rule	model separating it
ω↑=ω comm ₁	letter $\mapsto 2^{\text{depth}}$ in $(\mathbb{N}, +)$; shift doubles depths of 1-wire gates in List $\mathbb{N}$; shift = map suc
comm ₂ cong↑	the same, for 2-wire gates read gates only at depth ≥ 1 : no shift homomorphism
comm ₀	scalars in order, in List (Gate 0)



Scope today: the *structural* rules. No gate-specific rule set is shown minimal yet — but several were *reduced* by derivations (next slides).

Bian, Li, Ross, van de Wetering, Zhao, A
*Complete and Natural Rule Set for
 Multi-Qudit Clifford Circuits in All Odd Prime
 Dimensions* (QPL'26). Its verification is a
 separate paper in preparation.

- gates H , S (one qupit), CZ (two), any width n , any odd prime p ;
- read **modulo scalars**, the circuits present

$$\text{Pauli}(n) \rtimes \text{Sp}(2n, \mathbb{Z}_p);$$

- rules over p , a *primitive root* g , and $\frac{1}{2} \in \mathbb{Z}_p$;
- 16 rules in the paper (Figure 1); 15 here: $(SH)^3 = \varepsilon$ becomes a theorem.

```
data _SRel, _===_ : (n : ℕ) → WRel (Gen n) where
  order-S :      (1+ n) SRel, S ^ p === ε
  order-H :      (1+ n) SRel, H ^ 2 === XM_1
  M-power : ∀ k → (1+ n) SRel, XMg ^ k === XM (g ^ k)
  semi-MR :      (1+ n) SRel, XMg • S ^ (g * g) • Z ^ ((g * g + - g) * 1/2)
                  === S • XMg
  comm-HSHHS :   (1+ n) SRel, H • H • S • H • H • S === S • H • H • S • H • H
  order-CZ :      (2+ n) SRel, CZ ^ p === ε
  order-Ex :      (2+ n) SRel, Ex ^ 2 === ε
  comm-CZ-S↑ :    (2+ n) SRel, CZ • S ↑ === S ↑ • CZ
  semi-M↑CZ :     (2+ n) SRel, XMg ↑ • CZ ^ g === CZ • XMg ↑
  semi-Ex-S↑ :    (2+ n) SRel, Ex • S ↑ === S ↓ • Ex
  semi-Ex-H↑ :    (2+ n) SRel, Ex • H ↑ === H ↓ • Ex
  blake-c12 :     (2+ n) SRel, (S ^ p-1) ↑ • (S ^ p-1) ↓ • CX ^ p-1 • S ↓ • CX === CZ
  yang-baxter :   (3+ n) SRel, Ex ↑ • Ex ↓ • Ex ↑ === Ex ↓ • Ex ↑ • Ex ↓
  cz-slide :      (3+ n) SRel, Ex ↓ • Ex ↑ • CZ === CZ ↑ • Ex ↓ • Ex ↑
  semi-CX↑-CZ↓ : (3+ n) SRel, CZ ↓ • CX ↑ === CZ02 • CX ↑ • CZ ↓
```

ProjectiveClifford/Qupit/Paper-V1/Syntactics.agda (**qupit**), one
 comment elided

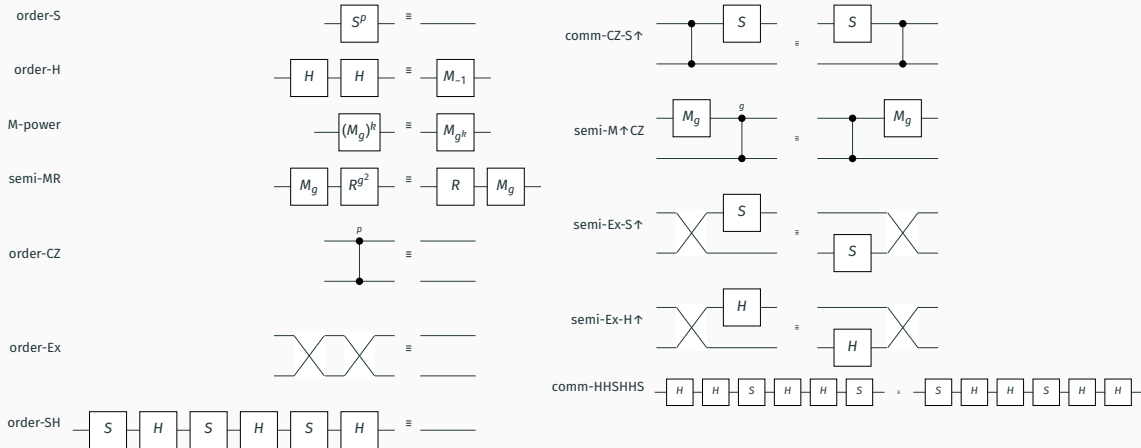
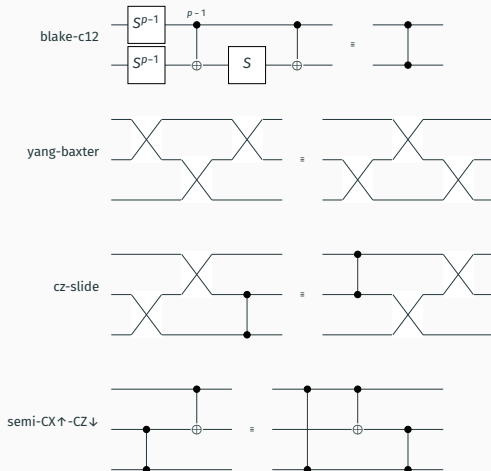


Figure 1 of the QPL'26 paper (the multiplier spelled over $R = S Z^{1/2}$); figures generated by Cir2Tikz from the Agda terms.



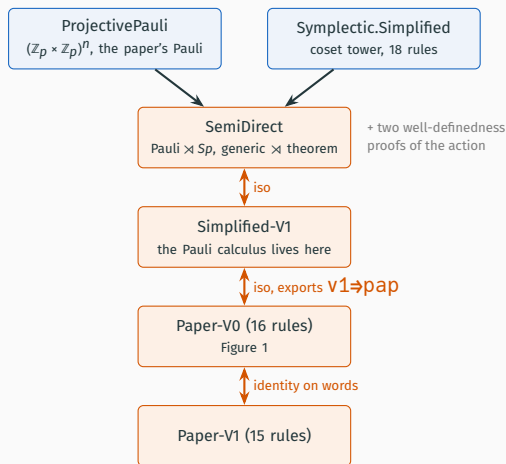
$Sp(2n, \mathbb{Z}_p)$ by a coset tower

(Examples/Groups/Symplectic, 35k lines):

- one level per wire, Selinger-style staircases as digits;
- one wire: $\mathbb{Z}_p \times \mathbb{Z}_p^* \times (\epsilon \mid HS^a)$, i.e. $p(p-1)(p+1) = |Sp(2, p)|$ normal forms;
- **67** $\rightarrow$ **18**: normalization uses 67 relations; a small set of natural ones derives them all.

-- *MainTheorems, Symplectic-Simplified-Theorems*
simplified-presentation :

$\forall n \rightarrow (n \text{ QRel, } _ == _) \text{ IsPresentationOf } (Sp\text{-group } n)$



Every layer above the base is **an isomorphism plus a composition**; no rule set re-proves what another has.

-- *MainTheorems, Qubit-Clifford-Theorems*

clifford-presentation :

$\forall n \rightarrow (n \text{ QRel, } _ == _) \text{ IsPresentationOf } (\text{Pauli} \ltimes \text{Sp } n)$

(PapSyn.Clifford-Relations._QRel, _ == _ in the source)

- reuses the paper's machinery verbatim: coset towers, the word-valued semidirect product, StarIsomorphism;
- derivability results: two Pauli-vs-CZ axioms dropped ($18 \rightarrow 16$); $(SH)^3 = \varepsilon$ a theorem in V1;
- $\approx 22k$ lines for the four rule sets.

The Clifford group is a **central extension** of the projective one by $\langle \omega \rangle$. Add ω , $\omega^p = \varepsilon$, centrality, and **correct** each relation by the scalar it picks up — only one needs it:

$$\text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \equiv \omega^{-1/8}$$

($\frac{1}{8}$ the inverse of 8 in $\mathbb{Z}_p$)

Clifford/Qubit: presentation-exact, against the central extension $A \rtimes_{\gamma} H$ with γ the Weyl cocycle $\frac{1}{2}$ sform; new ForStdlib: CentralExtension, FactorSetExtension (Rotman 9.8), CentralProduct.

Qubits, after Selinger (LMCS 2015), ProjectiveClifford/Qubit, Clifford/Qubit:

- Figure 8 modulo scalars presents $Sp(n)$ (symplectic map + $\mathbb{Z}_4$ phase function); $\cong$ the Clifford action on signed Paulis;
- an **extension** presentation $1 \rightarrow \mathbb{Z}_8 \rightarrow \text{Exact}(n) \rightarrow \text{CMS}(n) \rightarrow 1$, isomorphic to Figure 8;
- that ω has order exactly 8 needs a model: $2^n \times 2^n$ matrices over $\mathbb{Z}/17$ with $i = 4$, $\frac{1}{\sqrt{2}} = 14$, $\omega = 2$ — every relation by refl on tries.

A first, small **matrix** semantics — next: real ones.

The paper's stated limitation: its $U_3(\mathbb{Z}[\frac{1}{2}, i])$ and Clifford+ T studies stop at syntax, because the rings and unitary matrices were missing. **v-office** supplies them:

- rings $\mathbb{Z}[\frac{1}{2}]$, $\mathbb{Z}[\frac{1}{2}, i]$, $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ and matrices from **EucDomain** — an Agda port of Ross–Selinger's newsynth (17.6k lines);
- generators act as **two-level** matrices:

$$X_{[j,k]} = \begin{pmatrix} 1 & & & \\ & 0 & & 1 \\ & & 1 & \\ & 1 & & 0 \\ & & & & 1 & \\ & & & & & & 1 \end{pmatrix}$$

$X_{[j,k]}$

$$K_{[j,k]} = \begin{pmatrix} 1 & & & \\ & Y & & Y \\ & & 1 & \\ & Y & & -Y \\ & & & & 1 & \\ & & & & & & 1 \end{pmatrix}$$

$K_{[j,k]}$

$$i_{[j]} = \begin{pmatrix} 1 & & & \\ & i & & \\ & & 1 & \\ & & & 1 & \\ & & & & 1 & \\ & & & & & & 1 \end{pmatrix}$$

$i_{[j]}$

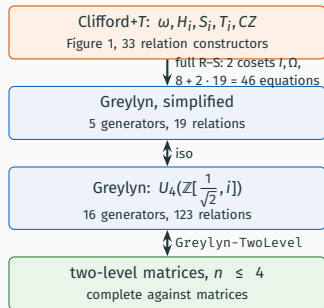
$$Y = \frac{1}{1+i}$$

Bian–Selinger (QPL'21), **every** n , against matrices:

```
-- Presentation.agda: the relations present  $U_n(\mathbb{D}[i])$ 
presentation : ( _==_ {n} ) IsPresentationOf (U n)
-- MainLemma.agda
relations-complete : {u v : Word (Gen n)} →
    [ u ]m ≡ [ v ]m → u ≈ v
-- Soundness.agda: decided on concrete  $4 \times 4$  matrices
rel-174 : [ l17 ]m ≡ [ r17 ]m
rel-174 = refl
```

- the normal form is **exact synthesis** (Algorithm 2.10), by well-founded recursion on (pivot, lde, #odd entries);
- soundness: commutations at any indices; the rest pulled back to dimension ≤ 4 and decided by `refl`;
- Greylyn's $U_n(\mathbb{Z}[\frac{1}{\sqrt{2}}, i])$ by the same code, generic in the ring: complete for $n \leq 4$.

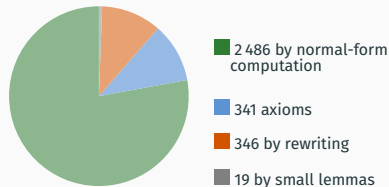
2-qubit Clifford+T (Bian–Selinger, QPL'22), end to end:



```

-- Clifford+T-2qubit/Presentation.agda:
--  $[-]$  is a group monomorphism into  $U_4(\mathbb{Z}[1/\sqrt{2}, i])$ 
presentation : Rel IsSubPresentationOf U 4
    
```

3-qubit Clifford+CS (QPL'23), in progress: the two-level $U_8(\mathbb{Z}[\frac{1}{2}, i])$ side is done; Step 4 of the R–S chain derives each of **3 192** relations from **508**:



Presentation/Tactics: $a \Gamma \vdash w \equiv v$
 view of our congruence, and the original paper's R–S theorem **as an instance** of Normalization.Reidemeister-Schreier.
 Step4/Main: 10 min, 2.2 GB.

A. Clément, arXiv:2602.06644: a 13-page paper, 264 pages of appendix.

data Gate : $\mathbb{N} \rightarrow \text{Set}$ where

H-gate : Gate 1

Z-gate : Gate 1

CZ-gate : Gate 2

CH-gate : Gate 2

Ex : Circuit $(2+n)$ -- Equation (8): the swap is a definition

Ex = CZ • H↓ • H↑ • CZ • H↓ • H↑ • CZ • H↓ • H↑

23 rules: (1)–(19) of Figure 4 minus (8), and the 5 swap rules of Remark 1; a sample:

data _SRel, _===_ : $(n : \mathbb{N}) \rightarrow \text{WRel} (\text{Gen } n)$ where

order-HZ : $(1+n)$ SRel, $(H \bullet Z) \wedge 8 \text{ === } \varepsilon$

merge-CZ : $(2+n)$ SRel, CZ • °CZ === Z↓

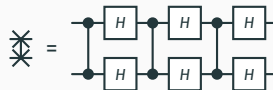
slide-CH : $(2+n)$ SRel, CH • H↑ • Z↑ • CH • H↑
 === H↑ • Z↑ • CH • H↑ • CH

square-CCZX : $(3+n)$ SRel, $(CH \downarrow \bullet CZ_{20}) \wedge 4 \text{ === } CZ \uparrow$

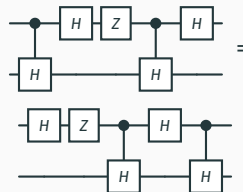
conj-PP : $(3+n)$ SRel, CZ↑ • PP↓ === PP↓ • CH↑

box-Z : $\forall k \rightarrow (1 + (4+k))$ SRel, Z↓ • $\bigwedge (4+k) \text{ === } \bigwedge (4+k) \bullet Z \downarrow$

the swap, by definition (8):



rule (11) slide-CH:



wire 0 is the bottom wire; CH is controlled by wire 1.

(19) is a **schema**: one rule per width ≥ 5 , about the multi-controlled box $\bigwedge$ of Definition 2.4, defined by recursion.

Gate matrices are taken $\times \sqrt{2}$, so entries lie in $\mathbb{Z}[\sqrt{2}]$; the power of $\frac{1}{\sqrt{2}}$ is kept apart:

$Op : \mathbb{N} \rightarrow \text{Set}$

$Op\ n = \text{Bits}\ n \rightarrow \text{Bits}\ n \rightarrow \mathbb{F}$

$Scaled : \mathbb{N} \rightarrow \text{Set}$

$Scaled\ n = \mathbb{N} \times Op\ n \dashv\dashv (\ell, M) \text{ reads } (1/\sqrt{2})^\ell \cdot M$

$\sim : Scaled\ n \rightarrow Scaled\ n \rightarrow \text{Set}$

$s \sim t = ((\sqrt{2}^\ell \text{proj}_1\ t) \cdot \text{proj}_2\ s) \doteq ((\sqrt{2}^\ell \text{proj}_1\ s) \cdot \text{proj}_2\ t)$

$$\frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

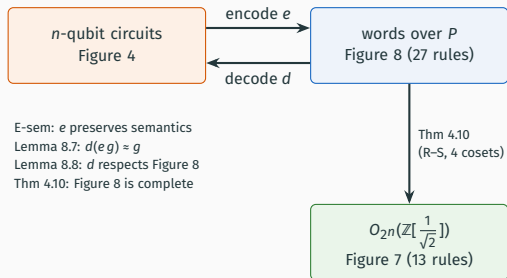
$I \otimes H$ as $(1, M)$, M over $\mathbb{Z}[\sqrt{2}]$

- matrices are computed on stored **tries** $\text{Mat}\ k$;
- soundness** (Prop. 3.1): each relator checked *once*, by `refl` on tries at the width it is drawn on; `localise` / `by-relator` lift it to *every* width;
- the schema (19) at every width by a symbolic lemma: the box is $(\frac{1}{\sqrt{2}})^\ell$ times a diagonal sign.

axiom-sound : $\{w\ v : \text{Circuit}\ n\} \rightarrow$
 $n\ \text{VRel}, w \text{ === } v \rightarrow [w] \sim [v]$

`Real-Clifford+CH/Semantics/Scaled.agda,`

`Soundness.agda`

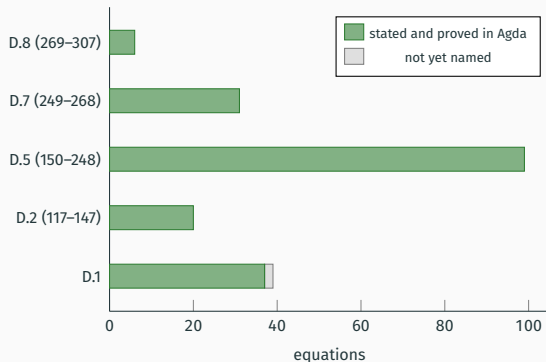


E-sem: e preserves semantics
 Lemma 8.7: $d(eg) \approx g$
 Lemma 8.8: d respects Figure 8
 Thm 4.10: Figure 8 is complete

Thm 4.4: complete — imported
 by the paper (Fang–Heunen–Kaarsgaard)

width	status (v-office-Cli-CH, Sep 29)
0, 1	complete outright (1 qubit: 16-state automaton NF)
2	from Theorem 4.4 at $N = 4$; Lemmas 7.4, 7.5 proved
≥ 3	BackAndForth: Thm 4.10 proved (from 4.4); e 's semantics and Lemma 8.7 proved at every width; Lemma 8.8 proved at every width ≥ 5 ; open at 3, 4 qubits

The hypothesis list shrank over one week: Theorem–8–9
 (4.10, 8.7, 8.8) $\rightarrow$ ' (4.4, 8.7, 8.8) $\rightarrow$ " $\rightarrow$ "" (4.4, and 8.8 at 3 and 4 qubits). No postulates; -- safe.



D.8 counts named statements, proved inside the width induction from completeness one width down.

- 294 files, **77 193 lines**, 48 commits in ten days (Sep 19-29);
- an n -qubit derivation uses completeness **one width down** as a black box — the paper’s “Lemma 5.1” is our induction hypothesis;
- a 3-wire evaluation `Same u v` becomes a derivation (`SemanticSteps`), ≈ 1 s per gate;
- structure instead of pages: two lemmas “replace some sixty diagram steps”; (203) “replaces the paper’s three pages”;
- Theorem 4.10 is the library’s **generic** R-S engine: Figure 8 over Figure 7, 4 cosets, 13×4 obligations, uniformly via one embedding.

In the paper

- Appendix A.4.1 prints E^{1-x} ; a decision procedure (Checks8) shows (36)/(37) hold only the other way round, and *fail at every instance* as printed.
- Checks8 found the **missing distinctness side conditions** of (30), (34), (42), and the negation parity of W_1/W_2 .
- The figure in the proof of (216)–(218) does not parse: as drawn it is **not semantically valid**; replaced by a structural argument.
- Footnote 13's “constructed in a similar way” needed a generic construction (six-index words).

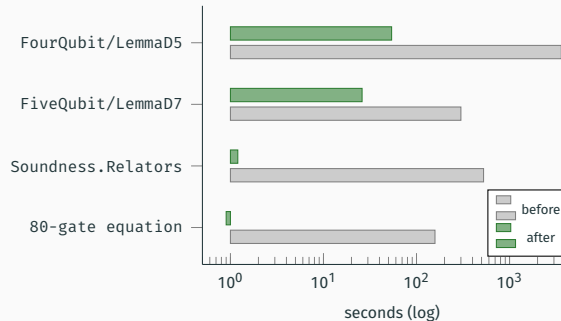
In our own transcription

- a swap network iterated the wrong way: invisible on 3 qubits, so (36)/(37) as transcribed were **unsound from 4 qubits on**; caught only when (65) was proved at symbolic width;
- (261) misread (a control dot) — the misread, harder statement was proved too; lesson: read the dots at 400 dpi;
- a draft typechecked while silently wrong: **zero** not imported, so it became a pattern variable — caught only by warnings.

Discipline: every claim tested numerically first (a pure-Python simulator), then decided by evaluation, then proved.

Operators are functions; Agda decides equality of two of them by applying and normalizing — for a product of forty gate matrices, a sum over every intermediate index, **exponential in the word**. It happens whenever two readings are not *syntactically* equal, for invisible reasons (3 vs $3 + 0$).

- `[[_]]` is **abstract**: readings compared by comparing words; defining equations exported as lemmas;
- $\sqrt{2}^n$ evaluated strictly level by level (lazily, $\sqrt{2}^{27}$ alone exhausts memory);
- never a lemma taking `Same (... u ...)` with a free word: pass a record (`Evaluated`) instead;
- a 4-wire evaluation is **not an option** (36 gates: no answer in 10 min) — use structure.



typechecking times reported in the commit history (“about an hour”,
“about five minutes”, “under a second” rounded)

Amy, Chen, Ross (QPL'17): gates ω , X , T , $CNOT$, $SWAP$; relations R_1 – R_{13} + symmetry laws. An operator is a **phase polynomial** $|x\rangle \mapsto \omega^{p(x)}|f(x)\rangle$, f affine:

$Op : \mathbb{N} \rightarrow Set$

$Op\ n = Bits\ n \rightarrow Bits\ n \times \mathbb{Z}_8$

- composition adds phases: **no sums**; each relation is refl on a table of ≤ 16 entries;
- soundness; 24 + 4 derived rules; **complete on one qubit** ($\omega^k T^a X^e$, 128 tables);
- n qubits: complete given the paper's imported lemmas (LaFont's affine normal forms, ...) as a record:

module Complete (NFs : **NormalForms** n) **where**

complete : $\forall \{w\ v : Circuit\ n\} \rightarrow [w] \doteq [v] \rightarrow n \vdash w \approx v$



$CNOT \cdot (I \otimes T) \cdot CNOT$: a diagonal, i.e. $f = id$, phase $p(x) = x_0 \oplus x_1$ — tables, not matrices

Same circuit layer, same congruence: ω 's centrality is the theorem `Central-Scalars.comm0`; 2829 lines; checks in under a minute.

Amy (QPL'18), *Towards large-scale functional verification of universal quantum circuits* — popl, 87k lines, **no assumptions**:

$$|x\rangle \mapsto 2^{-k/2} \sum_{y \in \mathbb{B}^m} e^{2\pi i P(x,y)} |f(x,y)\rangle$$

- exact amplitudes in $\mathbb{Z}[\zeta] = \mathbb{Z}[X]/(X^H + 1)$; rules [Elim], $[\omega]$, [HH], [Case];
- a **cost model**: each algorithm written once in a cost monad; value correctness + polynomial cost:

```
record Corollary-4-4 {n : ℕ} (C : Circuit n) : Set where
  field
    decides : value (decidec C) ≡ true ⇔ [ C ] ≈ idPS
    polynomial : cost (decidec C) ≤ decideBound n (length C)
    volume : cost (decidec C) ≤ volumeBound (n * length C)
decideBound-def : ∀ n ℓ → decideBound n ℓ ≡ 313 * (3 + (n + ℓ)) ^ 12
```

- Section 5.2 families for *every* size: QFT, n -bit Toffoli, Maslov, adder, hidden shift; Table 2 rows reproduced exactly (e.g. Toffoli₅₀: 190 paths, T-count 665);
- footnote 2's reduction from UNSAT, as an equivalence;
- beyond Clifford: normal forms **not unique**; a complete but exponential decision procedure.

Errata in the paper, each with a checked counterexample

- Lemma 4.2 needs Q odd at some input ($Q = 2x_1$ fails);
- Prop. 2.14's bound is $\max(2, k)$, not k ;
- Prop. 2.7: well-formedness is not preserved by composition.

independent of the presentation library: a second style of completeness (decision by rewriting, not normal forms of a group)

	POPL submission (tag, 22k lines)	since (other branches, up to 194k lines each)
core	words, inductive congruences, NF bundles, by-normalization, R-S engine, towers, $\times$ $\rtimes$ $*_M$ ext., circuits (3 structural rules)	0-ary gates, Central-Scalars, 4 structural rules; independence of axioms; central / factor-set extensions; generic circuit towers
symmetric & friends	S_n (2 semantics), $\mathbb{Z}_n$, trivial, $\mathbb{Z}_N \wr S_n$, Pauli	reorganized; projective Pauli
Clifford	qubit / qutrit Clifford+T (1 qubit), $U_3(\mathbb{Z}[\frac{1}{2}], i)$ — syntactic	qubit Clifford (all n , odd p), $Sp(2n, \mathbb{Z}_p)$, Selinger's qubit Clifford, exact versions with scalars
matrices	— (stated limitation)	$U_n(\mathbb{Z}[\frac{1}{2}], i)$ all n ; $U_{\leq 4}(\mathbb{Z}[\frac{1}{\sqrt{2}}], i)$; 2-qubit Clifford+T; Real-Clifford+CH over $\mathbb{Z}[\sqrt{2}]$; $\mathbb{Z}/17$ model
other semantics	—	phase polynomials (CNOT-dihedral); path sums with cost bounds
in progress	—	3-qubit Clifford+CS; Real-Clifford+CH Lemma 8.8 at 3–4 qubits; gate-specific minimality

Related work, lessons, outlook

Development	Prover	Circuits	Pres.	NF	Compl.
This library	Agda	✓	✓	✓	✓
Bian–Selinger developments	Agda	✓	✓	✓	syntactic
SQIR/VOQC, VyZX	Rocq	✓	–	–	sound only
QWIRE, CoqQ, Qbricks, QHLProver, IMD	Rocq/Why3/Isabelle	✓	–	–	–
Π as free rig groupoid	Agda (HoTT)	✓	categorical	✓	classical
$\sqrt{\Pi}$	Agda (partial)	✓	categorical	–	2q Clifford
mathlib group theory	Lean	–	✓	–	–
MathComp / Odd Order	Rocq	–	partial	–	–
Free groups (AFP), IsaFoR completion	Isabelle	–	✓ / –	✓	–

vs. mathlib: presentation *theory* (quotients, Coxeter); no infrastructure to prove a *given* relation set presents a *given* group — no coset enumeration, no verified transversals.

vs. rewriting: we never orient or prove confluence; a normal form is any verified *function* — sidestepping Squier’s obstruction (decidable word problem $\nRightarrow$ finite convergent presentation).

From the paper

1. **Setoids over syntax** beat quotients when you need to compute and to map *into* the syntax.
2. **Certificates, not search**: finite tables + five equations; the engine is proved once.
3. **Evaluation as a tactic**: map both sides to canonical data, compare by `refl` — no reflection needed.
4. **Index engineering**: left-biased successors make coverage a matter of unification.
5. **Constructions in pairs**: syntax theorem + semantic theorem, against `stdlib` bundles.

Since

6. **Tame the conversion checker**: abstract semantics, strict arithmetic, records instead of unfolded propositions.
7. **Simulate, decide, then prove at symbolic width**: a transcription bug invisible on three qubits surfaced only in a symbolic-width proof.
8. **Formalization finds errata**: in Clément's appendix and in Amy's path-sum paper — each confirmed by a decision procedure or a counterexample.
9. **Reuse compounds**: Theorem 4.10, the Clifford+ T 2-qubit proof and the qubit semidirect product all run on the paper's generic engine.

- **Upstreaming** the ForStdlib constructions (semidirect, amalgamated, central and factor-set extensions) to agda-stdlib.
- **Close the loops:** Lemma 8.8 on 3–4 qubits; Theorem 4.4 itself (a presentation of $O_N(\mathbb{Z}[\frac{1}{\sqrt{2}}])$, Fang–Heunen–Kaarsgaard); the remaining R–S steps of 3-qubit Clifford+CS.
- **Connect** the syntactic Clifford+T amalgams of the paper to the new matrix semantics.
- **Minimality** of gate-specific rule sets with the independence engine — e.g. Blake’s simplified near-Clifford presentations.
- The “264-page appendix” should become **ordinary, reusable, machine-checked mathematics**.

presentations as data

normal forms as bundles

coset tables as certificates

matrices as tries,
decided by `refl`

**complete rule sets,
machine-checked**

Thank you!

Questions?

Everything stated as a theorem is checked by Agda 2.8 + stdlib 2.4:

-- safe, no postulates; conditional results take their hypotheses as module parameters.